

PACKAGE INSERT

NEPHROGEN TABLET

Each tablet contains :-	
Vitamin B1 (Thiamine mononitrate)	100 mg
Vitamin B6 (Pyridoxine HCl)	200 mg
Vitamin B12 (Cyanocobalamin)	200 mcg

Product Description: A round, convex, off-white, film-coated tablet.

Pharmacodynamics:

Thiamine is an essential coenzyme for carbohydrate metabolism. Severe thiamine deficiency leads to the development of beri-beri characterized by peripheral neuritis, muscle wasting, muscle weakness, paralysis, cardiac failure, oedema and in severe deficiency, Wernicke's encephalopathy (demyelination of the CNS). Pyridoxine is involved in amino acid metabolism and also essential for the formation of haemoglobin and synthesis of GABA within the CNS. Pyridoxine deficiency leads to the development of peripheral neuritis. Cyanocobalamin is essential to growth, cell reproduction, hematopoiesis, nucleoprotein and myelin synthesis. Cyanocobalamin is converted to coenzyme B1 in tissues. Such is essential for conversion of methylmalonate to succinate and synthesis of methionine from homocysteine, a reaction that also requires folate. In the absence of coenzyme B12, tetrahydrofolate cannot be regenerated. Cyanocobalamin also may be involved in maintaining sulphydryl (SH) groups in the reduced form required by many SH-activated enzyme systems. Through these reactions, cyanocobalamin is associated with fat and carbohydrate metabolism and protein synthesis. Cyanocobalamin deficiency results in neurologic damage that begins with an inability to produce myelin and is followed by gradual degeneration of the axon and nerve head.

Pharmacokinetics:

Thiamine is well absorbed from the gastrointestinal tract following oral administration although the absorption of large doses is limited. It is widely distributed to most body tissue and it is not stored to any appreciable extent in the body and amounts in excess of the body's requirements are excreted in the urine as unchanged thiamine or as metabolites. Pyridoxine is readily absorbed from the gastrointestinal tract following oral administration. They are stored mainly in the liver where it is oxidised to 4-pyridoxic acid and other inactive metabolites which are then excreted in the urine. Cyanocobalamin substances bind to intrinsic factor and are then actively absorbed from the gastrointestinal tract. Cyanocobalamin is extensively bound to specific plasma proteins called transcobalamins. Transcobalamin II appears to be involved in the rapid transport of the cobalamins to tissues. It is stored in the liver, excreted in the bile and undergoes extensive enterohepatic recycling.

Indication:

As an adjuvant in the treatment of polyneuritis, neuralgia and diabetic neuropathy.

Recommended Dosage:

Adults: One tablet three times daily.
Tablet to be taken before or during meals.

Route of Administration:

Oral

Contraindications:

Hypersensitivity to any of the ingredients.
Vitamin B12 should not be used in patients with early Leber's disease.

Warnings and Precautions:

Patients with early Leber's disease (hereditary optic nerve atrophy) who were treated with Vitamin B12 suffered severe and swift optic atrophy. Hypokalemia and sudden death may occur in severe megaloblastic anaemia which is treated intensely with cyanocobalamin. Folic acid is not a substitute for cyanocobalamin although it may improve vitamin B12-deficiency megaloblastic anaemia. Exclusive use of folic acid in treating megaloblastic anaemia could result in progressive and irreversible neurologic damage. Hypokalemia and thrombocytosis could occur upon conversion of severe megaloblastic to normal erythropoiesis with cyanocobalamin therapy. Therefore, serum potassium levels and the platelet count should be monitored carefully during therapy. Cyanocobalamin or hydroxocobalamin should, if possible, not be given to patients without first confirming diagnosis and should not be used to treat megaloblastic anaemia.

Interactions with Other Medicaments:

Although the clinical importance is unknown, thiamine reportedly enhance the effect of neuromuscular blocking drugs. Pyridoxine reduces the effects of levodopa but this does not occur if a dopa decarboxylase inhibitor is also given. Many drugs may alter either the metabolism or bioavailability of pyridoxine such agents include isoniazid, penicillamine and oral contraceptive. Absorption of vitamin B12 from gastrointestinal tract may be reduced by aminoglycosides, aminosalicilic acid, anticonvulsants, biguanides, chloramphenicol, cholestyramine, cimetidine, colchicine, potassium salts and methyldopa. Serum concentrations may be decreased by concurrent administration of oral contraceptives. Many of these interactions are unlikely to be of clinical significance but should be taken into account when performing assays for blood concentrations.

Pregnancy and Lactation:

Nephrogen appears in breast milk. Very large doses have been reported to have a lactation-inhibiting effect. It is not known if Nephrogen can cause fetal harm when administered to pregnant women. However, pregnant and nursing mothers are advised to seek the advice of the physician before using the product.

Side Effects:

Adverse reactions seldom occur following oral administration of thiamine. Long-term administration of large doses of pyridoxine is associated with the development of severe peripheral neuropathies. It has been stated that this may occur with doses in excess of about 2 g daily. Administration of Vitamin B12 at doses greater than 10 µg daily may produce a haematological response in patients with folate deficiency; indiscriminate use may mask the precise diagnosis. Adverse reactions such as nausea, headache, paraesthesia, somnolence and urticaria were also reported. Erythema, desquamation of facial skin, rash and acne can be attributable to one of the vitamins.

Symptoms and Treatment of Overdose: Treatment is symptomatic and supportive.

Storage Conditions: Store below 25°C.

Pack Size: Blister pack: A box of 10 x 10, 50 x 10 and 100 x 10 tablets per strip.

Pack Size (export/tender only): A bottle of 1000 tablets.

Shelf-life: 2 years.

FURTHER INFORMATION CONCERNING THIS DRUG CAN BE OBTAINED FROM YOUR FAMILY PHYSICIAN / LOCAL GENERAL PRACTITIONER / PHARMACIST.

Manufacturer & Product Registration Holder:

SUNWARD PHARMACEUTICAL SDN. BHD.

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